

PICOPATT Internship

Quick Progress Update — May 2026

Explainable Clustering for Spatio-Temporal Picoclimatic Data

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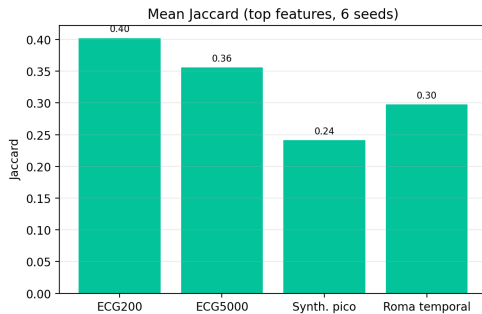
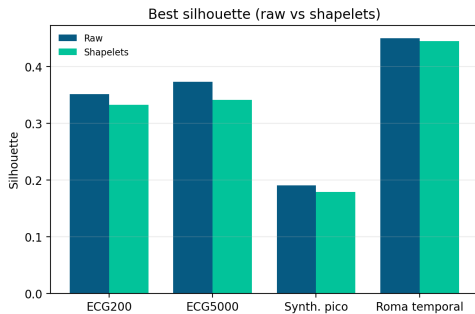
May 7, 2026

Snapshot

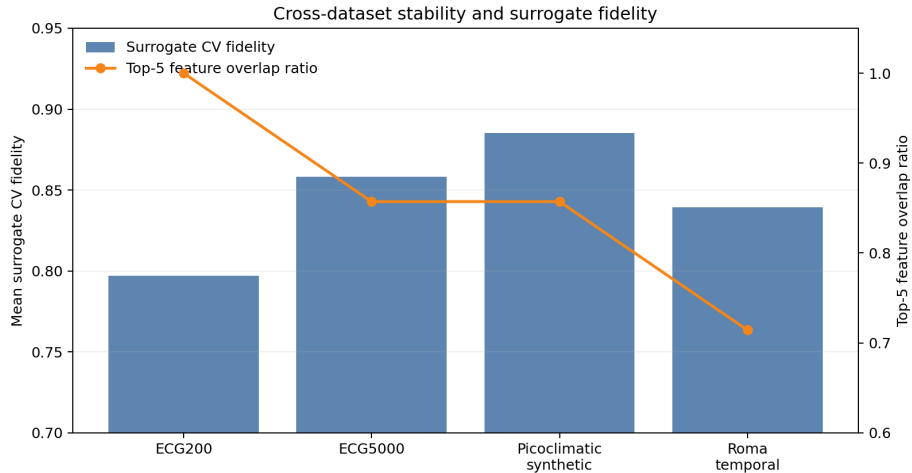
4 Proxy datasets ECG200, ECG5000, Synth. pico, Roma	6 Seeds for stability Feature overlap + fidelity	5 OK Research scripts (Apr 27) 0 failed; outputs archived	8 figs Exported visuals Slides + SHAP/LIME assets
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- Deck goal: less text, more evidence visuals.
- All numbers/figures sourced from archived runs under `outputs/`.

Cross-dataset summary



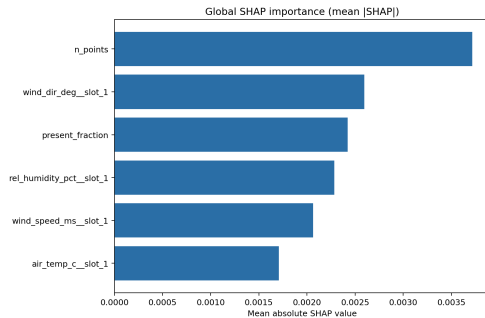
Stability evidence (6 seeds)



Explanation pipeline outputs (global)

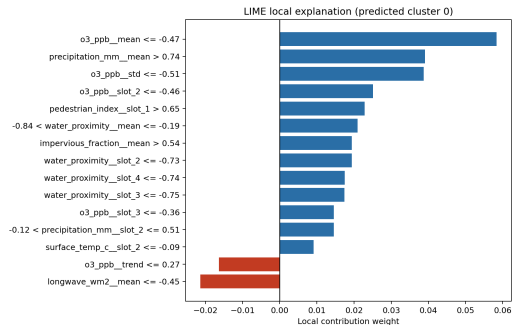


Pipeline overview (SHAP + LIME).

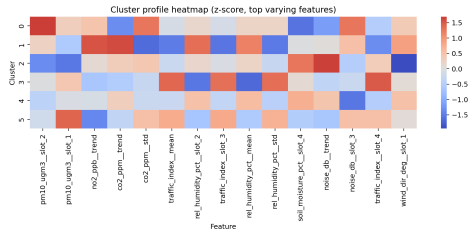


Global SHAP: top drivers.

Explanation outputs (local + cluster-level)

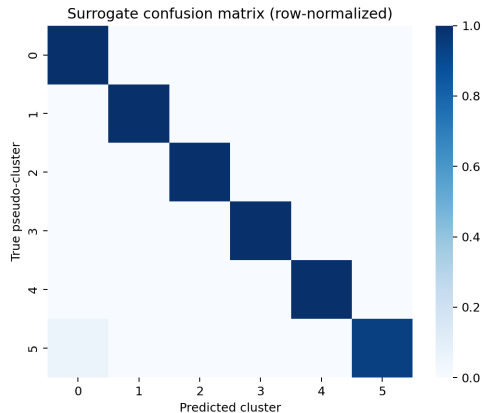


Local LIME explanation (example point).

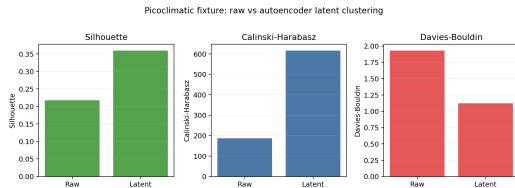


Cluster driver heatmap (surrogate-based).

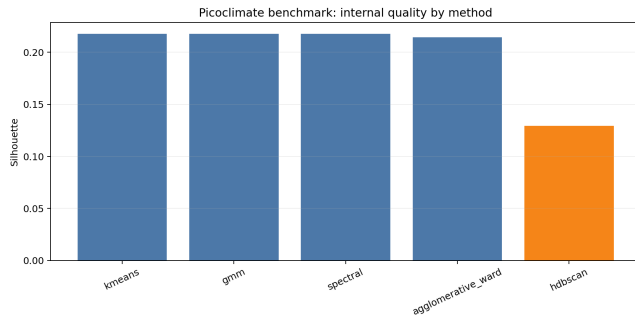
Surrogate quality + deep baseline



Surrogate confusion matrix.



Deep representation comparator.



Next (May)

- Freeze explanation schema (sensor-unit language).
- Expand benchmark matrix (spatial + deep baselines).
- Integrate real PICOPATT data when available.